

How to Use This Book

Book Structure

This book is organized into four parts that mirror a progressive learning path:

Part I: **Foundations of Generative AI** (Chapters 1-4) builds your conceptual base. You will understand how LLMs work, navigate the model landscape, make your first API calls, and learn where models excel and fail.

Part II: **Prompt Engineering** (Chapters 5-8) is the core craft. You will master prompt design, learn established techniques (zero-shot, few-shot, chain-of-thought), control output via API parameters, and build a disciplined iteration workflow.

Part III: **Conversational AI & RAG** (Chapters 9-11) moves from single prompts to applications. You will build multi-turn chatbots, work with embeddings and vector databases, and construct a full Retrieval-Augmented Generation pipeline.

Part IV: **Production & Responsible AI** (Chapters 12-14) prepares you for the real world. You will address security, manage costs, handle errors gracefully, and build ethical, evaluated AI systems.

How to Read

- **Sequential readers:** Follow the parts in order for a structured learning path
- **Reference readers:** Jump to any chapter; each is self-contained with cross-references
- **Hands-on readers:** Focus on the labs in the appendices alongside each chapter

Source Code & Companion Repository

Throughout the book, you will see callouts pointing to starter code, full exercises, and solutions. These references direct you to the book's companion repository:

[building-with-llms-companion](#)

For example, when exploring a chapter, you might see a tip like this:

 Tip

Starter Code: The companion repository contains full exercises, starter code, and solutions for making basic LLM API calls and building simple CLI Q&A scripts.

- [building-with-llms-companion/exercises/ch03/hello_llm.py](https://github.com/Kangsk/Kangsk.github.io/blob/main/building-with-llms-companion/exercises/ch03/hello_llm.py)
- [building-with-llms-companion/exercises/ch03/cli_qa.py](https://github.com/Kangsk/Kangsk.github.io/blob/main/building-with-llms-companion/exercises/ch03/cli_qa.py)

All labs, demos, and the capstone project are available in this companion repository. See **Appendix C** for setup instructions.

Additionally, we have compiled all high-resolution architecture and matrix diagrams referenced throughout the book into **Appendix E**. These diagrams are hosted in the companion repository's `diagrams/` folder.